

August 20, 2026

Dr. Bo Li
Editor-in-Chief
IEEE Internet of Things Journal

Dear Dr. Li,

Thank you for the decision on IoT-66559-2026, “FedKAN-IDS: Heterogeneity-Robust Federated Kolmogorov–Arnold Networks for NetFlow-Based IoT Intrusion Detection,” and for the opportunity to revise. We are grateful to both reviewers, whose comments were specific enough to be acted on rather than merely acknowledged.

The title has changed, to “What Federated KANs Actually Buy: Learning-Rate Robustness at a Compute Cost for NetFlow-Based IoT Intrusion Detection.” The previous title promised heterogeneity robustness; the revised measurements do not support that, and a title should say which of the two a reader is being handed.

We should say at the outset that this is not the same paper with corrections. Reviewer 2’s seventh comment asked whether the learning rate had been tuned per architecture. It had not: the submitted study used $\eta = 10^{-2}$ for every model. We therefore swept the learning rate over six values for all four architectures on all ten seeds, and the result changed our conclusions rather than confirming them. The parameter-matched MLP’s best learning rate is an order of magnitude away from the one we had used, and giving it that learning rate removes most of the accuracy gap and most of the variance gap we had reported as architectural.

Rather than restate a weakened version of the original claim, we have rewritten the paper around what the measurements now support. The system, the datasets, the partitioning protocol and the implementations are unchanged. The claims built on them are not, and the revised manuscript reports smaller numbers than the original in the places where the original was generous to our own method. Specifically: the headline binary advantage on NF-BoT-IoT-v2 falls from +5.49 to +0.76 pp once the baseline is tuned; the $2.6\times$ variance-reduction figure is withdrawn, since it is $0.71\times$, $0.33\times$ and $0.87\times$ on the other three cells and becomes $1.05\times$ on NF-BoT-IoT-v2 itself once both architectures are tuned; and the multi-class result is now reported as a range, +0.64 to +1.80 pp depending on the summary statistic, rather than as +1.735 pp at $p = 0.012$ from a single round.

We also owe the Editor two corrections that neither reviewer raised and that we found while acting on their comments.

First, a result that did not follow. Corollary 2 of the submitted manuscript concluded that FedKAN is intrinsically less sensitive to client heterogeneity. It does not follow from Theorem 1: the heterogeneity term in that bound applies identically to both architectures, and the spline-approximation residual is an additional error term for the KAN rather than a mechanism protecting it. The corollary asserted its conclusion instead of deriving it. We have removed it.

Second, a reporting choice that overstated precision. All headline accuracies in the submitted manuscript were taken from the final communication round. We measured the

round-to-round oscillation over the last ten rounds and it is 2.1 to 2.7 percentage points across all four architectures, which is larger than one of the effects we reported as significant. Every accuracy in the revised manuscript is the mean over the final five rounds, and we state the estimator explicitly.

Single-machine re-run. Every table in the revised manuscript is now generated from runs executed on one machine. The headline accuracy table and its last-round companion were regenerated from scratch rather than carried over, because the submitted version’s figures came from an RTX 4090 while the new experiments ran on CPU, and a table mixing the two would not support the comparisons we draw from it. The generating script prints the device of every run it reads and refuses to proceed silently if more than one appears. The submitted numbers came from an RTX 4090 and a mixture of hardware would have made the new and old numbers incomparable. We also report inference and training cost on that hardware, which speaks to the deployment questions both reviewers raised.

Marked-up copy. We enclose a clean manuscript and a marked-up copy in which new text is highlighted. We should be straightforward about one limitation. Where a passage was rewritten rather than edited, an automatic comparison cannot always place the deleted text in context, so parts of what was removed do not appear in the marked-up copy at all. We measured this rather than leave it implicit: of the previous manuscript’s body sentences, 34.6% are gone and 55.4% of the present one is new, but the automatic comparison marks only 57% of the new sentences and 45% of the deleted ones. The blue marking is reliable where it appears; its absence does not mean a passage is unchanged. We therefore list the substantive deletions explicitly in the response letter rather than leaving the reader to infer them from the markup.

Data and code. The simulator, all model implementations, the added baselines, every experiment reported here, and the scripts that regenerate each figure and table are available at <https://github.com/haodpsut/FedKAN-IDS-v2>.

We hope the reviewers find the changes responsive. Where we have declined a suggestion we say so and explain why rather than passing over it in silence.

Sincerely,

Phuc Hao Do, on behalf of all authors